

JOSHUA POLLICK

Graduate passionate about creating, modding and playing games. I also have experience with web development. I'm a great team player as well as an excellent communicator. I'm eager to put my skills to the test and help make games I can be proud of.



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University of Warwick 2025 - 2026

Games Engineering (MSc): Advanced Computer Graphics; Games Engineering; Games Engine Design and Development; Fundamentals of Games Research, Development and Management; Innovative Simulation Design and Development

University of Cambridge 2019 - 2022

Computer Science (MA): Advanced Graphics and Image Processing; Advanced Operating Systems; Advanced Computer Architecture; Information Theory; Computer Networking; Concurrent and Distributed Systems; Databases

Lancaster Royal Grammar School (A-Levels 2017 - 2019)

A*/A*/A*/A* - Computer Science / Maths / Further Maths / Physics

Education

Fusion Esports Centre: Playtester for unreleased game (2026)

- Tested and evaluated game mechanics, flow and UX.
- Provided precise feedback and bug reports

Triage App Programmer (2023-2024)

- Worked with a junior doctor to produce a prototype app for assisting with triage in ER & GP waiting rooms
- Identified inefficiencies and redundancies from patient and doctor perspectives
- Modelled problems with use of surveyed data and expert opinions to identify how an app could alleviate issues
- Iteratively refined the UX with feedback from professionals
- Created using Flask

Eggcup: Inventory Developer & Server Maintainer (2015 - 2021)

- Eggcup is a surplus food charity. As it scaled, it required a Food Inventory that could deal with irregular stock, and also required minimal training for use by volunteers.
- Created an internal web-app using Django, Javascript and MySQL. It provided an easy, familiar interface for stock input/output; item normalisation to coalesce identical items; a flexible overview to help determine what food should be used when; and customizable statistical summaries for management to provide to sponsors.
- Refined the app over time, combining written feedback with in-person usability testing, further tailoring it to the charity's unique needs.
- Provided maintenance and support over SSH and in-person, for the server device, inventory app and wordpress website.

University of Glasgow: Multimodal Interaction Group intern (2018)

- Developed a proof-of-concept VR car demo, driven by "physically" interacting with the steering wheel via VR controllers rather than joysticks, to assist the group's research into motion sickness.
- Wrote easy to understand public summaries of research papers for the group website.

Work

Languages

- C++ • Java • HTML
- C# • HLSL • Javascript
- Lua • SQL • Python

Skills

- Communication +
- Teamwork +
- Group Management
- Project Management

Tools/Frameworks

- Unreal +• DX12 • Winsock
- Unity • Qt • Harmony
- ImGui • Git • Django
- FMOD • Mirror • Netcode (Unity)

Projects This is a highlight of the programming projects I've worked on, with a focus on recent games and game-related projects. More details (and more projects) can be found at my [portfolio](#). Self-started projects are marked with **blue bullet points**.

- Group 2026** • **Skyfire Uprising**: 43-person group project, in teams of 6. I implemented a data-driven dynamic enemy spawning system; a waypoint system; designed the level layout; built a significant portion of the playable level; took on a group leader role; and co-ordinated with other teams to ensure consistency.
- 2025** • **Frantic Pinball**: Entry for [46 hour "Frantic Game Jam"](#), made in Unity with 3 others. I had the original idea; implemented the player controls and puck behaviour; "polished" and presented the final build.
- 2024** • **"Artifaction"**: WIP game, made in Unity with 1 other developer. I've worked on every part, focusing on networking; modular base systems; and integrating external APIs (Vivox and Steam Audio).
- 2021** • **SpellCast**: 2D 3v3 proof of concept game, made in Unity with 2 others. I created the base systems (frontend and backend) used throughout the game. Entry for [NSE Games Innovation Challenge 2021](#).
- **Riffr**: 6-person real-time collaboration web-app. I worked on the front-end: I designed the main page; implemented the UI; integrated the other members' components into the UI.
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Solo 2026 • **"RogueFrame"**: TODO

- **Ray Tracer**: CPU-based path-tracer implementing Oren-Nayar; mirrors; glass with refraction; conductors and dielectrics with single-scattering GGX micro-facet; single-scattering layered BSDFs with homogenous participating media and Beer's Law; tile-based multithreading; binned SAH BVH building; Intel® Open Image Denoise; importance sampling weighted by P/d^2 ; multiple importance sampling; light tracing; instant radiosity.
 - **Slay the Spire 2 Mods**: One fixed a race condition with steam cloud syncing, patched after my report. The other enables users to have multiple simultaneous run save files. Implemented in C# with Harmony and Godot.
 - **Chatroom**: C++ chatroom with a server-client architecture, implemented using winsock, Dear ImGui and FMOD. Gracefully handles concurrency between different clients as well as the frontend/backend
- 2025** • **JSON Parser**: C++ JSON Parser with support for comments, capable of reading/writing to files
- **DX12 Renderer**: C++/HLSL DX12 rasteriser with alpha testing; vertex shader animation; mesh instancing; third-person controls; animations with a state-machine controller; OBB collisions using SAT; velocities and inelastic collisions; normal mapping; deferred rendering; point and spot lights with shadow mapping, directional lights with cascading shadow maps; Cook-Torrance PBR with Lambert + Fresnel-Schlick + GGX.
 - **CPU Rasterizer**: Z-buffer rasteriser with back-face culling, linear attribute interpolation and Lambertian lighting, written in C++ with custom maths library.
- 2023** • **Arkshot Mods**: Enables network-synced customisation of gameplay values for the Unity game Arkshot. Written using BepinEx, with a mix of C# hooks and CIL transpilers
- 2021** • **Incremental Backup**: Fully modular and generic incremental backup engine implemented in Java.
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Other Experience and Achievements

- **Netcraft AQA A Level Computing Award**: Top 50 results in AQA Computer Science
- **Programming Society**: Co-founded, led and ran a school society dedicated to providing extra-curricular python teaching, as well as assisting student hobby projects
- **Mentoring**: Provided mentoring to a Year 8 student, overcoming their struggles in Computer Science & Physics at LRGS, volunteered in an online programming teaching community
- **Developments in VLSI Design Talk**: Researched, prepared & presented a 25 minute talk on very large scale integration in microprocessors for Churchill College, University of Cambridge. Received an award for "an outstanding talk"
- **Overwatch Commentator at Cambridge vs Oxford e-sports Varsity 2023-2024**: <https://youtu.be/JvjkFmFBgZ4> and <https://www.twitch.tv/videos/2127447965>. Quickly summarized live information and provided analysis in-person and on-stream
- **Totally Normal High School**: Play-tested and balanced mechanics for Cambridge University Roleplaying and Tabletop gaming Society's in-house TTRPG
- **Vecter**: Worked with the developer by providing: numerous bug reports; and feedback which helped steer the direction of the game. I also later moderated the community discord server. I was the first to break multiple milestones in the game, and currently hold the world record in "Speed Demon"
- **Roblox**: Worked on Roblox projects in Lua around 2014, in particular a "Minigames" game.